Timor Digital 2032

# Timor Digital 2032

Timor-Leste's National Strategic Plan for Digital and ICT Development

ENGLISH VERSION

Timor Digital 2032

**Led by TIC Timor, Public Institute**

TIMOR Tecnologia da Informação e Comunicação

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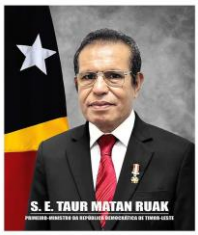
# FOREWORDS

## From the Prime-Minister of Timor-Leste

### His Excellency, Taur Matan Ruak




As the Prime Minister of Timor-Leste, considering it is critical for nation-building to have a strategy for developing digital and information and communication technologies (ICTs), it is my pleasure to announce that the government has completed the development of Timor Digital 2032.

In today's society, all plans aimed at national development can benefit from the opportunities offered by digitisation and ICT. Government and every area of human activity can benefit from it. We can only expect that digital technologies and ICT will progress to be at the forefront of every domain, enabling the transformation of all of society's activities. Any government development plans and aspirations are incomplete without considering the impact, contribution, drive and need to develop digital and ICTs.

With the leadership of TIC Timor I.P., the government has developed a strategy that leverages digital and ICTs to not only improve service delivery and backend administration to develop a closer relationship with its citizens but also to develop the areas with more significant impacts on human development and to progressively support all of society's growth.

The government's vision for Timor Digital 2032 is to develop a society that can thrive internally by ensuring the benefits of digitisation are being passed on to all citizens. It aims to achieve this by ensuring digital and ICTs serve as a mechanism for delivering public services and enhanced access to markets, personal development, and job opportunities to those who would otherwise be excluded. Furthermore, through digitisation the government seeks to empower its citizens to participate in public policy, adding and generating value for themselves and the Timorese society. Ultimately, Timor Digital 2032 will yield a citizenry poised to fully take part in the upcoming global and digitised economy and society.

The success of Timor Digital 2032 will depend on the strategy remaining a political priority, maintaining alignment with government development priorities, and staying relevant to the nation. Therefore, all stakeholders must closely align with this strategy to ensure coordination and collaboration at the political, policy, and implementation levels.

Therefore, I invite all to work together toward the government's vision of a digital society for Timor-Leste and to welcome Timor Digital 2032.

## From the Minister of the Presidency of the Council of Ministers

**His Excellency, Fidelis Manuel Leite Magalhães**

The impetus for the Government of Timor-Leste to develop a national strategy for digital and ICTs grew out of the Public Sector Reform program, which identified the need to plan for the use of digital and ICTs in improving government service delivery, backend administration, transparency, public participation, and social inclusion.

This need resulted in the development of a Ten-Year Strategy: Timor Digital 2032. This strategy presents a framework to implement digitisation and ICTs in the most impactful areas for the government: service delivery, human development, and the development of Timor-Leste as a nation, such as governance, economy, education, health and agriculture while also allowing adaptability along with its implementation.

The strategic plan creates mechanisms to enable senior coordination at the strategic level to ensure it stays relevant, a priority of the Government of Timor-Leste and aligned with national development priorities and plans. It also creates implementation coordination mechanisms that count on the participation of the civil society in acceleration and innovation labs and the design of services to ensure a citizen-centric government. For all these reasons, we envision that Timor Digital 2032 will be a success and mark the start of a new phase in government service delivery for Timor-Leste.

## From the TIC Timor Executive Director and President of the Board

**Roberto Caetano De Sousa Vicente**

Digital and ICTs underpin all aspects of government and society. As the national ICT agency of Timor-Leste responsible for implementing ICT policy and development strategy, as well as the platforms necessary for the implementation of e-Government, TIC Timor has led the development of a strategic plan which will allow Timor-Leste to fully use digital and information technology to improve the way the government functions internally, to advance the quality and reach of government service delivery, and to a broader extent, to develop a Timorese society that is better equipped for a digital future.

Along with leading the design of the Timor Digital 2032, TIC Timor is already well underway in its implementation and preparation. TIC is working across all pillars of the strategy and modernising the government's ICT infrastructure, with plans to significantly improve existing government networks and datacentres. Currently, TIC Timor is connecting all municipalities to the government network, establishing key cybersecurity practices by developing and planning a Government Security Operations Centre and Network Operations Centre, launching a Digital Unique Identifier (ID) and its Directorate, establishing a Digital Advisory Team, driving the drafting of enabling legislative framework, and working with the government, civil society, development partners, and other stakeholders to prepare for the delivery of the strategic goals. TIC Timor is ready to support the Government of Timor-Leste in implementing Timor Digital 2032 and for its vision to be a success.

# OBJECTIVES

Timor Digital 2032 is a result of the VIII Constitutional Government of Timor-Leste's Public Administration Reform Program and is a Ten-Year Strategic Plan to develop digital and ICTs in Timor-Leste with an emphasis on their application in areas that are critical and have the most significant impact on human and economic development such as government service delivery (e-governance/ digital government, inclusive economy, health, education, and agriculture). To stay aligned with the government's development priorities, the plan is also prepared for adaptability and for additional areas to be included progressively.

The objective of the Ten-Year Strategy is to establish a framework that will enable Timor-Leste to successfully develop digital and ICTs by focusing on their application in key areas: the digital strategic pillars (key sectors for human and economic development); the enabling pillars (areas to be developed for successful implementation, technology access, adoption, and value creation); and the strategic principles (fundamental principles to be accounted for at all times).

Equally important in developing Timor Digital 2032 will be the fact that it will provide a reference and common understanding to all and will lay the basis for all stakeholders to work together towards common and clear goals related to digital transformation. The strategy also proposes coordination and implementation mechanisms that will ensure alignment with the government's development priorities and maximisation of benefits and value for money for the investments in digital technologies and ICT.

# VISION

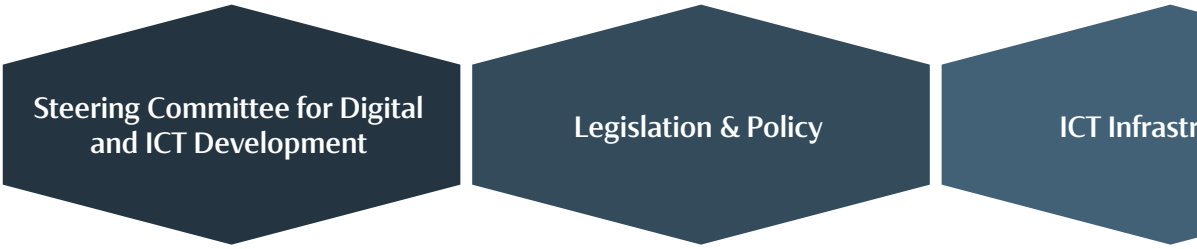
**Government:** With the help of digital and ICTs, achieve a digital government that is simple, transparent, and easily accessible to all, one where the government is not only present on mobile phones and computers but everywhere, at the sucos (villages), schools, clinics, and enables and fosters civic participation in public policy. In partnership with the civil society, develop a public service that takes into consideration the needs and desires of the population.

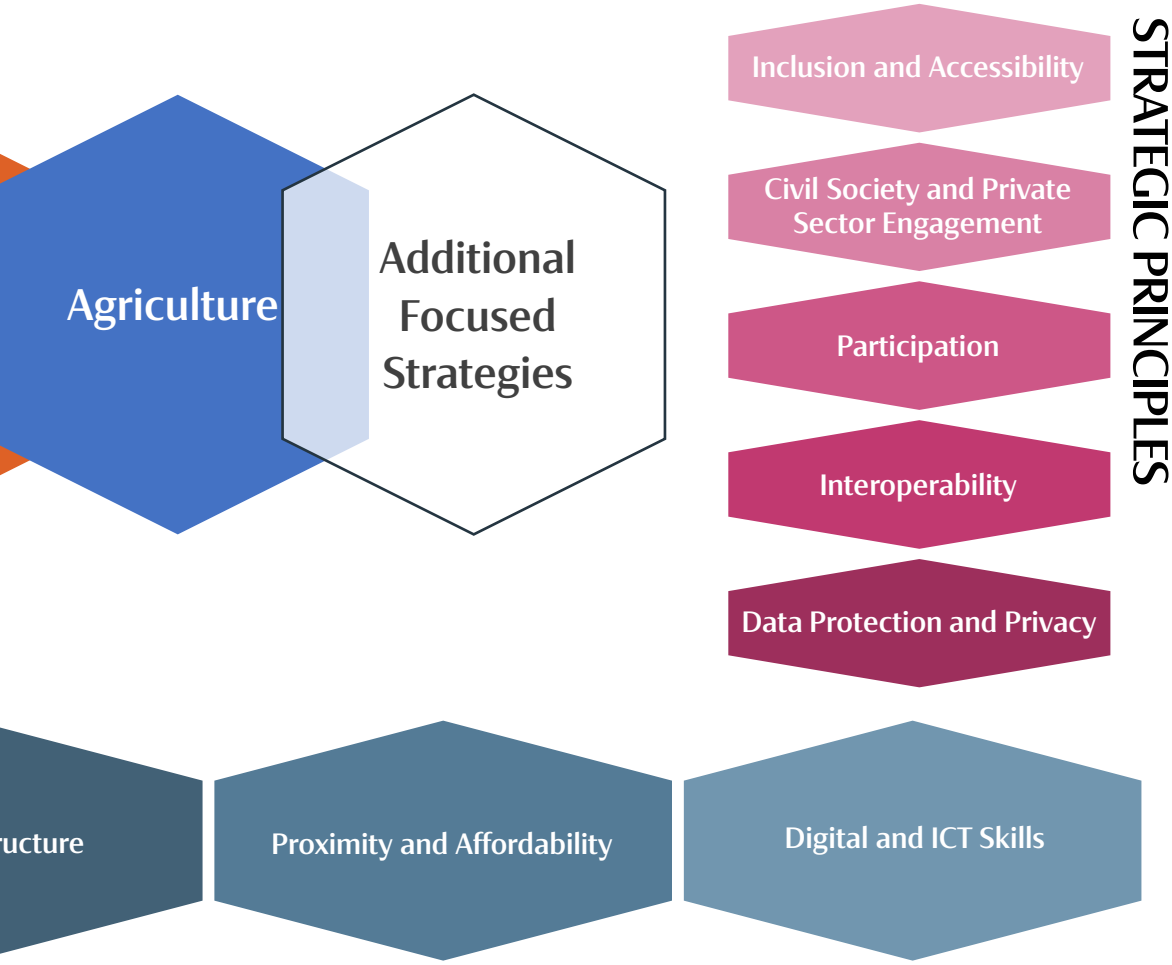
**Society:** Develop and achieve a society where all can create and share value and contribute to the Digital Age and development of a Digital Economy, where the young generation will in the future be able to work online, not only within Timor but without borders to compete in a global market; and the elderly, the people in rural areas, people living with any form of disability, or the vulnerable are not excluded from government services or from participating in markets, work and development opportunities in the anticipated fully digitalised economy.

# STRATEGIC FRAMEWORK

## STRATEGIC DIGITAL PILLARS

## ENABLING PILLARS




STRATEGIC PRINCIPLES
Inclusion and Accessibility
Civil Society and Private Sector Engagement
Participation
Interoperability
Data Protection and Privacy
Additional Focused Strategies
Agriculture
ructure
Proximity and Affordability
Digital and ICT Skills

# STRATEGIC DIGITAL PILLARS

- The Strategic Digital Pillars in the framework are the core elements of Timor-Leste's National Strategic Plan for Digital and ICT Development, and each pillar means the government will develop a focused strategic plan that includes an implementation roadmap on how Digital and ICTs will be applied to the areas of governance, inclusive economy, health, education, and agriculture.
- Each Strategic Digital Pillar will result in one focused strategic plan. Each plan should be updated regularly during the ten-year implementation.
- All plans must address how the strategic principles will be ensured when applicable.
- To maintain alignment with the government development priorities, the steering committee will progressively, during the ten years, identify additional strategic digital pillars, such as justice, social inclusion, youth employment, investment and trade, and tourism.

## Governance

**Strategic Objective:** Achieve a government that is simple, transparent, and easily accessible to all, one where the government is not only present on mobile phones and computers but everywhere, at the sucos, schools, clinics, and enables and fosters civic participation in public policy.

- The Governance Strategic Digital Pillar is the most important of all pillars as the initiatives in this pillar will enable the digital development of all other sectors.
- The need to use digital and ICTs to improve the government backend and service delivery was identified in the Public Administration Reform Program which determined important to develop an implementation plan focused on the use of digital and information and communication technologies to improve governance, transparency, accessibility and reach, a plan that progressively digitalises government services, streamlines and simplifies citizens and business interactions with the government, modernises public sector service delivery, improves government back-office, effectiveness, accountability, and enables and fosters participation in public policy.
- To develop a governance-focused strategic plan, TIC Timor will need to assess all government information management systems, to develop a catalogue of all government services, and to develop an interoperability architecture and policy which uses the Unique ID as the authentication mechanism to enable interoperation and simplification of government services. To develop a citizen-centric digital government this plan needs to be developed in close collaboration with civil society organisations and count on cross-ministry collaboration.
- Access to government services for the citizens living in the last mile is a government priority, hence it is important to connect all administration offices to the government ICT network and to establish a one-stop shop in all sucos: *Uma Digital* to serve as a government ServiceDesk and link citizens and local government to the central government, NGOs, Development Partners, markets, and private sector stakeholders.

- Simplicity, transparency, accountability and participation are key elements of a modern digital government, needs also identified by the Civil Society, therefore, is important to centralise all government websites and services in a single government internet portal, including government structure, services, news, budget execution, reports, tenders, vacancies, and public policy consultation as well important is to apply the same look and feel to all government services.
- To coordinate the management of the strategic plan at the policy-making level, the Government of Timor-Leste will establish a Steering Committee for Digital and ICT Development, having as members the ministers from key ministries, representatives from the civil society and the private sector.
- At the working level, TIC Timor will establish a Digital Advisory Team which will coordinate the development and execution of implementation plans to deliver the strategic plan, support government ministries and agencies in their digital transformation journey, and work with the civil society and private sector.
- Also at the working level, all government ministries and agencies should establish Digital Acceleration Teams to work with TIC Timor Digital Advisory Team and the civil society to leverage digital and ITCs to improve their back-office and service delivery.
- The development of digital and ICT competencies for the whole of the government and public service is key to the success of the strategic plan, from the policy-making to the operational level. Therefore, TIC Timor will establish a Digital Acceleration and Capabilities Development Centre which will serve as a think-tank and collaboration platform for GovTech and digital and ICT development matters for the government, civil society, and the private sector to work together to lead the implementation of the strategic plan and develop solutions that are truly citizen and business-centric as well as to train policymakers, ICT technical staff, civil society and public servants.

## Inclusive Economy

**Strategic Objective:** With the help of digital and ICTs, enable a digital economy from which all people can benefit, including people living with any form of disability, minorities, and people in rural areas, and ensure all have enhanced and expanded access to markets with the help of digital and ICTs.

- Develop a focused strategic plan for an Inclusive Digital Economy that will bring new opportunities and enhance access to markets and support to women and men, the elderly, people in rural areas, and people living with any form of disability. The plan should include a national e-commerce strategy, promote mobile payments, and provide solutions to enhance access to markets and the promotion of local products and tourism.
- Develop a Suco and product promotion portal: Portal Merkado Bo'ot to collect information from farmers, artisans, and others and promote products and tourism from rural and last-mile areas.

## Health

**Strategic Objective:** Develop specific solutions to improve the Health Sector service delivery quality and reach across Timor-Leste by leveraging the opportunities of Digital and ICTs.

- Develop a focused strategy for the Health Sector that includes connecting all government health care providing facilities to the government ICT network, leveraging on the Unique ID implementing individual patient management systems.
- Equip health-providing facilities with digital technologies such as video-conferencing and others to enable areas such as telemedicine and remote support and to foster national and international collaboration.

## Education

**Strategic Objective:** Develop specific solutions to improve the education service delivery quality and reach across Timor-Leste by leveraging the opportunities brought by Digital and ICTs.

- Develop a focused strategic plan and roadmap for the education sector that plans for the use of digital and ICTs in education. The plan should include connecting all education facilities to the government ICT Network and to the internet, the installation of ICT labs, the development of digital and online libraries, and knowledge repositories, and using the Unique ID to implement student records management systems, and academic management systems for the whole sector.

## Agriculture

**Strategic Objective:** Develop specific solutions to improve support for the agriculture sector in production and access to markets across Timor-Leste by leveraging the opportunities offered by Digital and ICTs.

- Develop a focused strategic plan for the agriculture sector which leverages on the use of Digital and ICTs for production, and monitoring, facilitates access to national and international markets and plans for the use of e-Agriculture tools.

## Additional Focused Strategies

**Strategic Objective:** Leverage the opportunities of Digital and ICTs to progressively develop further focused strategic plans to maintain alignment with GoTL development priorities.

- Progressively develop additional focused strategies for additional sectors aligned with the government's priority development areas and reforms. These can be sectors such as justice, social inclusion, youth employment, investment and trade, and tourism.

# ENABLING PILLARS

The Enabling Pillars are important to ensure effective and efficient delivery of the strategy and the best return on the investments. These pillars will be the object of several master plans and roadmaps to progressively and continually develop all enabling pillars over the ten-year period.

## Steering Committee for Digital and ICT Development

The Steering Committee for Digital and ICT Development will be the high-level coordination body for Timor Digital 2032

- Establish a policy-making level Steering Committee for Digital and ICT Development led by PCM, with representation from key ministries and government agencies, the Civil Society, and the Private Sector.

## ICT Infrastructure

The ICT infrastructure enabling pillar focus on developing the national ICT infrastructure and the government ICT infrastructure.

- The digital transformation will increase an already existing demand to develop a highly skilled and specialised ICT workforce for the government, and the need to develop and maintain a sustainable and secure government ICT. The government will thus apply a one-government ICT approach and fully centralise the ICT and digital technologies in TIC Timor for all ICT staff, ICT policy, digital government, networks, datacentres, systems administration, corporate applications, cybersecurity, systems, and applications development.
- To ensure investments in digital and ICTs are consolidated, aligned and provide the best value-for-money the government will develop oversight and coordination mechanisms for its ICT spending and investments.
- To keep its ICT infrastructure modern and increase its reach the government will progressively and continuously upgrade its infrastructure, datacentres, and systems, and will extend its network to reach and connect all government administration, sucos, schools, and health-providing facilities.
- Fully establish the Unique ID and register the entire population to serve as the authentication mechanism for accessing government services.
- Enhance national broadband networks and invest in improving the quality and cost of Internet connectivity.
- The government will develop a digital and ICT investment attracting strategy in order to create an ecosystem of innovation, enable partnerships and foster entrepreneurship in digital technology, ICT, and telecommunications sectors.

**Proximity and Affordability**

Develop a government, with the help of Digital and ICTs that is close to people, everywhere, on their phones and computers and in schools, clinics, and sucos. Develop mechanisms to improve internet affordability.

- Implement mechanisms to improve physical proximity and affordability for citizens and businesses: free or low-cost data access to portal and applications.
- Establish government services presence across several mediums, such as: one-stop-shops in all Sucos: Uma Digital: One-stop-digital-shop and ServiceDesk to access Government Services and link citizens to markets, NGOs, and local government.
- Centralised Government Portal and Centralised Digital Government Services approach. Applications with explanation videos and narration for accessibility.

**Legislation and Policy**

Develop the enabling legislation for digital development while ensuring that the principles of privacy and protection of citizens are taken into consideration when undertaking the Timor-Leste digital transformation journey.

- Continue developing enabling legislation and policies such as DPP Law (in progress), Cybercrime Law (in progress), E-Commerce Law (in progress), Intellectual Property Code (in progress), UID Decree-Law (in progress), and further policies and data sharing agreements

**Digital and ICT Skills**

Develop a society that is not only ready for the Digital Age but can benefit and contribute to it and drive and sustain the national digital transformation.

- Partnering with universities, development partners, and others, establish a Digital Acceleration and Capabilities Development Centre to train policy makers and ICT technical staff. Organise training programs and fellowships for policymakers, ICT technical staff, and citizens to gain the digital and ICT skills needed to create and foster a digitally enabled environment.
- Equip citizens with digital and ICT skills to enjoy the benefits and contribute to a digital Timor-Leste and digital society by updating the education curriculums for all levels.
- Equip the ICT workforce at TIC Timor with a range of skills such as digital transformation, systems development, cybersecurity, datacentres and cloud management, and computer networking to be able to foster and support the digitisation of the Government of Timor-Leste.
- Develop a comprehensive National Digital and ICT Skills development plan.

# STRATEGIC PRINCIPLES

The Strategic Principles are principles to take into consideration in all government plans in what regard to digital development and each plan should address how the principles will be assured when applicable.

### Inclusion and Accessibility

- All government strategic plans, roadmaps, master plans, and initiatives should address the principle of Inclusion and Accessibility.
- All plans must include details on how Inclusion and Accessibility will be taken into consideration and ensured.

### Civil Society and Private Sector Engagement

- All government strategic plans, roadmaps, master plans, and initiatives should address how civil society and the private sector can be involved, engaged and supported when applicable.
- Whenever possible, the civil society and the private sector should be involved in the development and implementation of the goals in the strategic plan.

### Participation

- All government strategic plans, roadmaps, master plans, and initiatives should address the principle of participation.
- All plans must include details on how participation will be enabled and fostered.

### Interoperability

- All government strategic plans, roadmaps, masterplans, and initiatives should address the principle of Interoperability between government systems to ensure de-duplication of processes, required documents, and inputs from citizens and businesses.
- All plans must include details on how systems will interoperate.

### Data Protection and Privacy

- All government strategic plans, roadmaps, master plans, and initiatives should address the principle of Data Protection and Privacy when applicable.
- All plans must include details on how Data Protection and Privacy will be taken into consideration and ensured.

# STRATEGIC GOALS SUMMARY

**Goal 1** – Establish a Policy-Level Steering Committee for Digital and ICT Development led by the PCM and key ministers, civil society, and the private sector as members.

**Goal 2** – Develop digital sector-focused strategic plans for governance, inclusive economy, education, health, and agriculture.

- **Goal 2.1** - Develop a governance-focused digital development strategic plan.
- **Goal 2.2** - Develop a digital development strategic plan for an inclusive economy.
- **Goal 2.3** - Develop an education-focused digital development strategic plan.
- **Goal 2.4** - Develop a health-focused digital development strategic plan.
- **Goal 2.5** - Develop an Agriculture-focused digital development strategic plan.

**Goal 3** – Progressively develop focused strategic plans for additional sectors in alignment with the Government of Timor-Leste's Development Priorities, such as inclusion, youth employment, climate change, justice, tourism, and fisheries.

**Goal 4** – TIC Timor to establish a Digital Advisory Team to support GoTL Ministries in the digital transformation journey at the working level.

**Goal 5** – All government agencies to establish Digital Acceleration teams in to work with TIC Timor Digital Advisory Team and leverage digitisation and ICTs to improve service delivery.

**Goal 6** – Establish a Digital Acceleration and Capabilities Development Centre to serve as a think-tank, acceleration and collaboration platform and to train policymakers, ICT technical staff, Civil Society, Private Sector and Public Servants.

**Goal 7** – Fully centralise government ICT in TIC Timor for all government ICT matters such as human resources, IT networks, datacentres, systems administration, ServiceDesk, email and corporate applications, systems and applications development, and cybersecurity.

**Goal 8** – Develop coordination mechanisms to consolidate and align government spending and investments in Digital and ICTs.

**Goal 9** – Centralise all government websites and services in a single government internet portal, including government structure, services, news, spending, reports, tenders, vacancies, and public policy consultation as well, apply the same look and feel to all government digital services.

**Goal 10** – Establish a one-stop shop in all sucos: Uma Digital to serve as a government ServiceDesk and link citizens and local government to the central government, NGOs, Development Partners, markets, and private sector stakeholders.

**Goal 11** – Enhance national broadband networks and invest in improving the quality and cost of Internet connectivity.

**Goal 12** – Progressively and continuously upgrade the government ICT infrastructure, datacentres, and systems, and extend the government ICT network to reach and connect all government administration, sucos, schools, and health-providing facilities

**Goal 13** – Update educational curriculums for all levels to enable the youth to develop digital and ICT skills, starting at preschool.

**Goal 14** – Fully establish the Unique ID, register the entire population and progressively establish the UID as the authentication mechanism to access government services.

**Goal 15** – Implement a Suco and product promotion portal: Portal Merkado Bo'ot to promote products from rural and last-mile areas in national and international markets.

**Goal 16** – Develop a digital and ICT investment attracting strategy in order to create an ecosystem of innovation, enable partnerships and foster entrepreneurship in digital technology, ICT, and telecommunications sectors.

**Goal 17** – Develop a comprehensive National Digital and ICT Skills development plan.

**Goal 18** – Develop legislation and policies to enable all goals.

**Goal 19** – Merge TIC Timor I.P. with DNIC (the Directorate National of Information and Communication Technology) from MTC into a new Ministry for Digital and ICT Development.

Governance · Inclusive economy · Health · Education · Agriculture · Additional focus areas
Digital and ICT Skills · Legislation & Policy · ICT Infrastructure · Proximity and Affordability
Steering Committee for Digital and ICT Development

# MAPPING OF EXPECTED IMPACTS AND BENEFITS

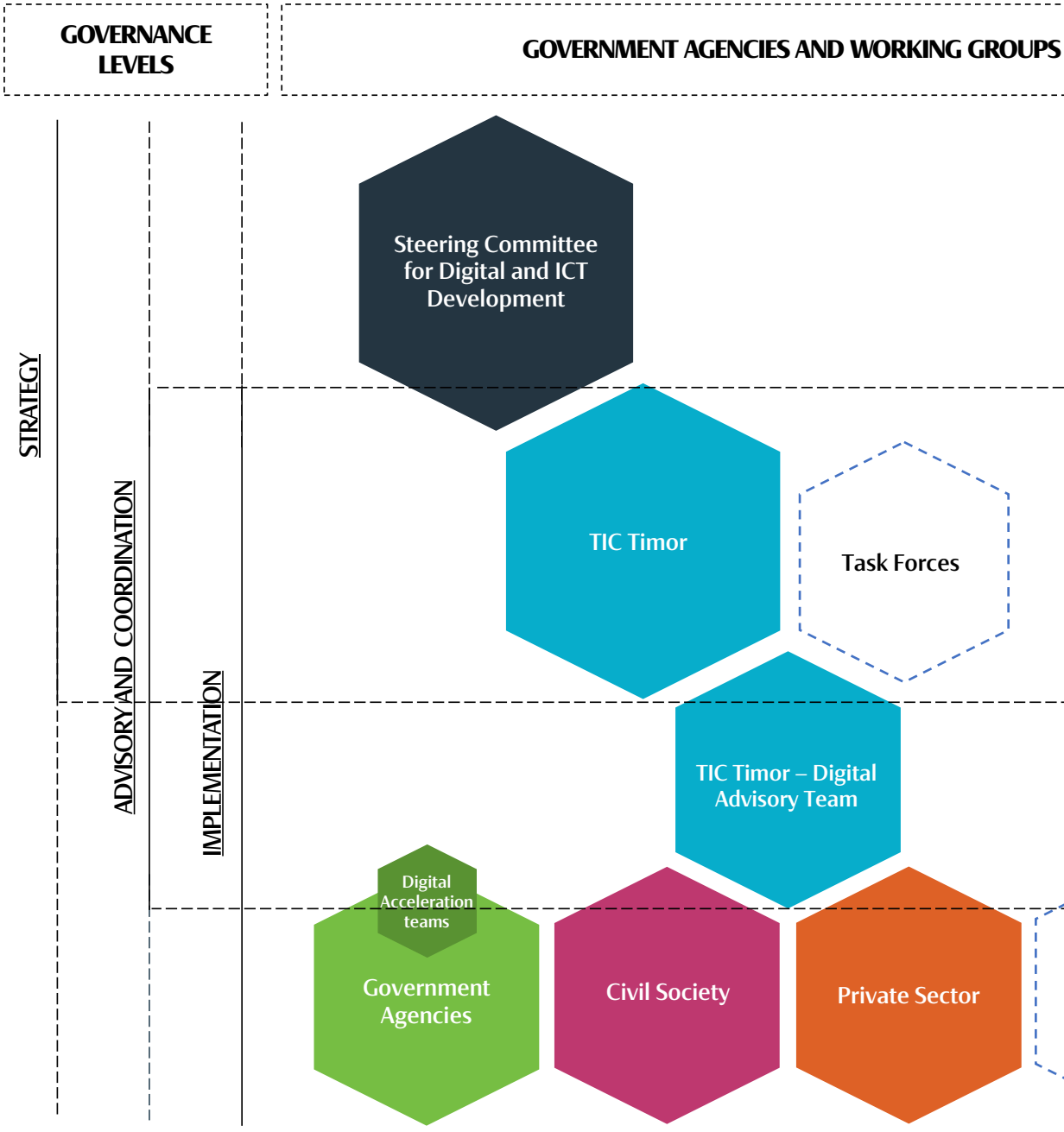
| Goals | Impact and benefits |
|---|---|
| **Goal 1** – Establish a Policy-Level Steering Committee for Digital and ICT Development. | Establishing a policy-level Steering Committee for Digital and ICT Development, with key ministers, civil society, and the private sector as members will ensure digital development matters relish the required high-level political commitment and support.<br>Establishing such a multi-stakeholder committee will enable cross-sectorial and whole-of-society policy coordination for all digital development matters, which will result in an improved engagement of citizens, businesses and other stakeholders, a faster digital transformation, better impact feedback, an adaptative implementation, while ensuring benefits are aligned and responsive to society's needs. Therefore, establishing the steering committee will result in government services and digital development that are tailored to society's needs and aspirations, and in improved social inclusion and accessibility. |
| **Goal 2** – Develop digital sectorial strategies for governance, inclusive economy, education, health, and agriculture. | |
| **Goal 2.1** - Develop a digital sectorial strategy for governance. | Developing a governance-focused strategic plan will increase government internal efficiency and productivity, and improve flows of information from citizens to government, government to citizens, and within the government itself.<br>Government digital transformation will enable collaboration among government departments and will result in reduced workforce costs. Digitalising and simplifying requirements for government services will improve the user experience for citizens and businesses, foster public participation and increase government transparency. |
| **Goal 2.2** - Develop a digital sectorial strategy for inclusive economy. | Digital technologies will assist the government in progressively developing a digital and inclusive economy, improve digital skills, improve access to markets, foster private sector development and create new jobs. |
| **Goal 2.3** - Develop a digital sectorial strategy for education. | Digital technologies will assist the government in progressively developing the Education sector, which will result in improved digital Skills, increased employability, an entrepreneurship environment and the overall improvement of the education sector. |
| **Goal 2.4** - Develop a digital sectorial strategy for health. | Digital technologies will assist the government in progressively developing the Health sector, and in improving citizens access to health care and the health sector efficiency. |
| **Goal 2.5** - Develop a digital sectorial strategy for agriculture. | Digital technologies will assist the government in progressively developing the Agriculture sector, improving farmers' digital skills, sector productivity, and access to markets, along with creating new jobs. |

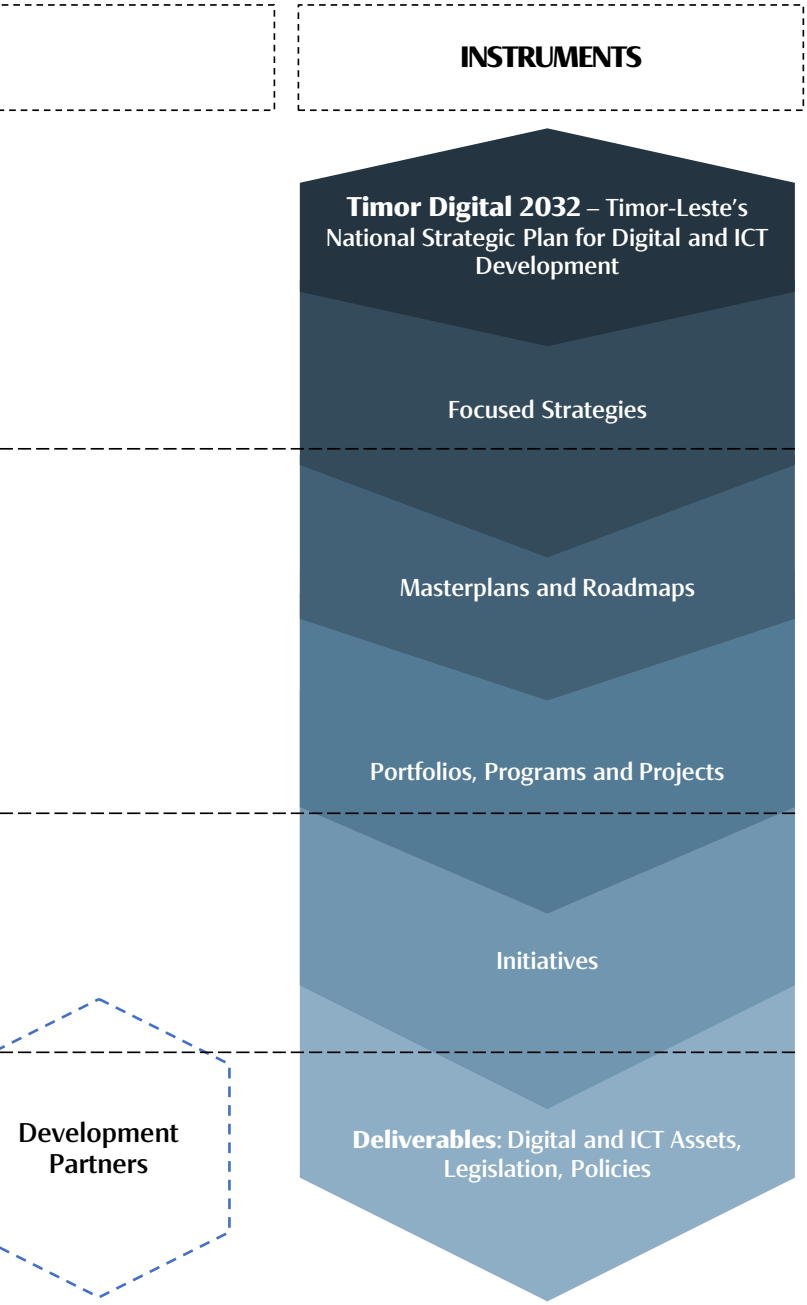
| Goals | Impact and benefits |
|---|---|
| **Goal 3** – Progressively develop additional sector-focused strategic plans in alignment with the Government of Timor-Leste's Development Priorities, such as inclusion, youth employment, climate change, justice, tourism, and fisheries. | Digital technologies will assist the government in progressively developing all sectors. |
| **Goal 4** – TIC Timor to establish a Digital Advisory Team to support GoTL agencies in the digital transformation journey at the working level. | Improved support to government agencies. Better alignment of government agencies' needs with digital technologies, resulting in improved efficiency and service delivery. Establishes entry points for implementation at the working level.<br>Improved digital skills across government agencies. |
| **Goal 5** – All government agencies to establish Digital Acceleration teams in to work with TIC Timor Digital Advisory Team and leverage digitisation and ICTs to improve service delivery. | Improved support to government agencies. Better alignment of government agencies' needs with digital technologies, resulting in improved efficiency and service delivery. Establishes entry points for implementation at the working level.<br>Improved digital skills across government agencies. |
| **Goal 6** – Establish a Digital Acceleration and Capabilities Development Centre to train policymakers, ICT technical staff, Civil Society, Private Sector and Public Servants and to serve as a think-tank, acceleration and collaboration platform. | This will enable the development of solutions and delivery of results better aligned with whole-of-society needs, that are tailored to society and resulting in improved Citizens', Businesses', and stakeholders' engagement, inclusiveness and accessibility and digital skills for policymakers, ICT technical staff, Civil Society, Private Sector and Public Servants. |
| **Goal 7** – Fully centralise government ICT in TIC Timor for all government ICT matters such as human resources, IT networks, datacentres, systems administration, ServiceDesk, email and corporate applications, systems and applications development, and cybersecurity. | The centralisation of the government ICT will enable the government to develop a highly specialised ICT workforce and infrastructure which will have a wide range of benefits such as:<br>• Improved government service delivery.<br>• Increased Government Services availability.<br>• Improved alignment of government needs with ICT and Digital Technologies.<br>• Improved cybersecurity.<br>• Reduced capital investment and operational costs.<br>• Easier to implement a digital government.<br>• Improved trust in ICT by government agencies. |

| Goals | Impact and benefits |
| --- | --- |
| **Goal 8** – Develop coordination mechanisms to consolidate and align government spending and investments in Digital and ICTs. | This will result in improved value-for-money for the government investments in digital and ICTs and will ensure all government initiatives are aligned with the digital and ICT development strategic plan and are fit for the purpose. |
| **Goal 9** – Centralise all government websites and services in a single government internet portal, including government structure, services, news, spending, reports, tenders, vacancies, and public policy consultation as well, apply the same look and feel to all government digital services. | Centralising all government agencies' websites will offer simplicity in finding and accessing services and information which will result in better-informed citizens, improved access to government services, an improved user experience for citizens and businesses, simplified service requirements, and increased public participation and transparency. |
| **Goal 10** – Establish a one-stop shop in all sucos: Uma Digital to serve as a government ServiceDesk and link citizens and local government to the central government, NGOs, Development Partners, markets, and private sector stakeholders. | Developing a one-stop shop in all Sucos as a cross-stakeholder platform will provide better access to government services, and better support to communities. Therefore, resulting in a positive impact on communities' lives, improved social inclusion, and better access to health and education services.<br>It will as well bring new opportunities, as acquiring digital skills, improved access to markets, and support to family and women-owned enterprises. |
| **Goal 11** – Enhance national broadband networks and invest in improving the quality and cost of Internet connectivity. | Enhancing national broadband networks is necessary for better and more affordable access to government and mobile services. Access to technology, being a critical enabler of digital development, will pave the way for a wide range of benefits to society. |
| **Goal 12** – Progressively and continuously upgrade the government ICT infrastructure, datacentres, and systems, and extend the government ICT network to reach and connect all government administration, sucos, schools, and health-providing facilities. | Continuously upgrading the government ICT infrastructure will improve the efficiency and availability of government service delivery. |
| **Goal 13** – Update educational curriculums for all levels to enable the youth to develop digital and ICT skills, starting at preschool. | Enabling contact with digital and ICTs from a young age will result in a society with great Digital skills and further access to specialised jobs, therefore enabling the participation and development of a digital society. |

| Goals | Impact and benefits |
|---|---|
| **Goal 14** – Fully establish the Unique ID, register the entire population and progressively establish the UID as the authentication mechanism to access government services. | The Unique ID will result in increased government internal efficiency and productivity, and in improved flows of information from citizens to government, government to citizens, and within the government itself. The Unique ID enables the simplification of the requirements for government services, enables collaboration among government departments, and can result in reduced workforce costs. |
| **Goal 15** – Implement a product promotion portal: Portal Merkado Bo'ot to promote products from rural and last-mile areas in national and international markets. | The portal will enable Local and National promotion, thus having a positive impact on communities' lives<br>Will enable communication to have increased Access to markets and opportunities for family and women-owned enterprises. |
| **Goal 16** – Develop a digital and ICT investment attracting strategy in order to create an ecosystem of innovation, enable partnerships and foster entrepreneurship in digital technology, ICT, and telecommunications sectors. | Developing a strategy for the attraction of investment in digital technologies, ICT and Telecommunications will result in competitive services and prices, improved affordability and accessibility, new and improved digital services. It will help to develop a vibrant entrepreneurial and innovative private sector, create employment opportunities, provide economic empowerment for women and families, improve livelihoods and result in the economic development of the whole society |
| **Goal 17** – Develop a comprehensive National Digital and ICT Skills development plan. | A comprehensive plan for developing digital and ICT skills will result in several benefits:<br>▪ Improved digital skills for the whole society.<br>▪ Improved Government service delivery.<br>▪ Positive impact in communities' lives.<br>▪ Highly-Skilled Public Services Workforce. |
| **Goal 18** – Develop legislation and policies to enable all goals. | A fit-for-purpose regulatory framework will develop a catalysing and welcoming environment that supports investment, competition, and innovation while ensuring fundamental protections are in place: privacy to ensure personal data is protected, secure and used ethically and enable a thriving innovative and entrepreneurial environment and society. |
| **Goal 19** – Merge TIC Timor I.P. with DNIC (the Directorate National of Information and Communication Technology) from MTC into a new Ministry for Digital and ICT Development. | Improved alignment of all matters relating to digital transformation and development. |

# IMPLEMENTATION GOVERNANCE FRAMEWORK






The Implementation Governance Framework comprises three governance levels, from Policy Making and Strategy down to a Working Level. These are Strategy, Advisory and Coordination, and Implementation.

The figure shows which Government Agencies and working groups act at each level and which instruments are the responsibility of different levels, agencies, and working groups.

# GOVERNANCE LEVELS

## Strategy

- The Strategy Governance Level will include the Steering Committee for Digital and ICT Development and TIC Timor, both will be responsible for managing the Timor Digital 2032 strategy, establishing strategic guidance, and developing and approving all plans and policies resulting from the Timor Digital 2032 strategic plan.

## Advisory and Coordination

- The Advisory and Coordination Governance Level will be TIC Timor and TIC Timor Digital Advisory Team, which will work with government agencies, civil society, and the private sector in advocating for the use of digital and information and communication technologies to improve governance and service delivery and progressively streamline and simplify government interaction with citizens, business, and itself.
- The Advisory and Coordination Governance Level will be responsible for developing and managing masterplans, roadmaps, portfolios of programs and projects, and initiatives to achieve the strategic goals through tangible deliverables such as digital and ICT assets, legislation, and policies.

## Implementation

- The Implementation Governance Level will be composed of TIC Timor, TIC Timor Digital Advisory Team, government agencies, civil society, and the private sector and will be responsible for the delivery of digital and ICT assets, legislation, and policies.

# AGENCIES AND WORKING GROUPS ROLES

## 1 - Steering Committee for Digital and ICT Development

- Establishing the Steering Committee for Digital and ICT Development is the first goal of Timor Digital 2032 and the cornerstone of the Strategic Plan.
- Chaired by PCM and comprised of ministers from key ministries, heads of government agencies, and representatives from civil society and the private sector as consultative members. The proposed government members are the Minister of the Presidency of the Council of Ministers; the Minister of Communications and Transport; the Executive Director for TIC Timor IP; the Governor of the Central Bank of Timor-Leste; the Minister of Finance; the Minister of Education, Youth and Sports; the Minister of Higher Education, Science, and Culture; the Minister of Health; the Minister of State Administration; the Minister of Agriculture and Fisheries; and the Minister of Tourism, Trade, and Industry.
- Be the custodian of Timor Digital 2032 and act as the strategic decision board for national digital transformation matters. Ensure the alignment of development partners' initiatives in digital and ICT development with the strategic pillars, strategic principles, and enabling pillars of the strategic plan.
- Ensure cross-agency coordination and alignment of all government initiatives in digital and ICT development with the strategic pillars, strategic principles, and enabling pillars of the strategic plan.
- Enforce ongoing simplification and digitisation of the government services and interfaces with citizens and the private sector.
- Develop and apply continual service improvement strategies to government services, as well as progressively align and monitor alignment with the government's development priorities.
- Foster collaboration, institute peer support and establish task forces when needed.
- Revise and approve and oversee the development, implementation roadmaps, masterplans, and portfolios of initiatives resulting from the Timor Digital 2032 strategic plan.
- Prioritise the development of additional digital pillars for additional sectors such as justice, social inclusion, youth employment, investment and trade, and tourism.
- Perform an annual revision and publication of the Timor Digital 2032 strategic plan.

## 2 - Task Forces

The Steering Committee for Digital and ICT Development will establish task forces to develop and implement specific strategies, roadmaps, and initiatives when relevant. Members of these task forces will be the government agencies relevant for the tasks at hand.

## 3 - TIC Timor

TIC Timor is the lead implementing agency for the strategy as the national ICT agency responsible for implementing ICT policy and development strategy, as well as the platforms necessary for the implementation of e-Government.

## 4 - TIC Timor – Digital Advisory Team

TIC Timor will establish a Digital Advisory Team, responsible for working closely with all government agencies to help leverage digital and ICT use to improve GoTL service delivery.

## 5 - Government Agencies

The role of government agencies is to work with TIC and stakeholders to continuously leverage the use of digital and ICTs to improve service delivery to citizens, businesses, and other government agencies.

## 6 - Government Agencies – Digital Acceleration Teams

Each government ministry and agency will avail a team from the current structure to be dedicated to digital acceleration and to work with the TIC Timor and TIC Timor Digital Advisory Team.

## 7 - Civil Society

The civil society representative role is to represent Timor-Leste citizens and NGOs and ensure that Timor Digital 2032 is aligned with their needs. Civil society is a consultative member.

## 8 - Private Sector

The role of the private sector representation is to represent Timor-Leste private businesses and ensure that Timor Digital 2032 is aligned with their needs. The private sector is a consultative member.

# GOVERNANCE INSTRUMENTS

### Timor Digital 2032 Strategic Plan

Main governing document. All focused strategies should be developed and maintain alignment with the framework and its strategic pillars and principles.

### Focused Strategies

Focused sectorial strategic plans for each Strategic Digital Pillar and aligned with the government's development and reform priorities.

### Masterplans and Roadmaps

Breakdowns of focused strategies into high-level portfolios and steps to follow along with and high-level budget estimates.

### Portfolios, Programs and Projects

Detailed, actionable, and initiative-oriented portfolios of programs and projects with technical specifications and exact budget costs and timelines.

### Initiatives

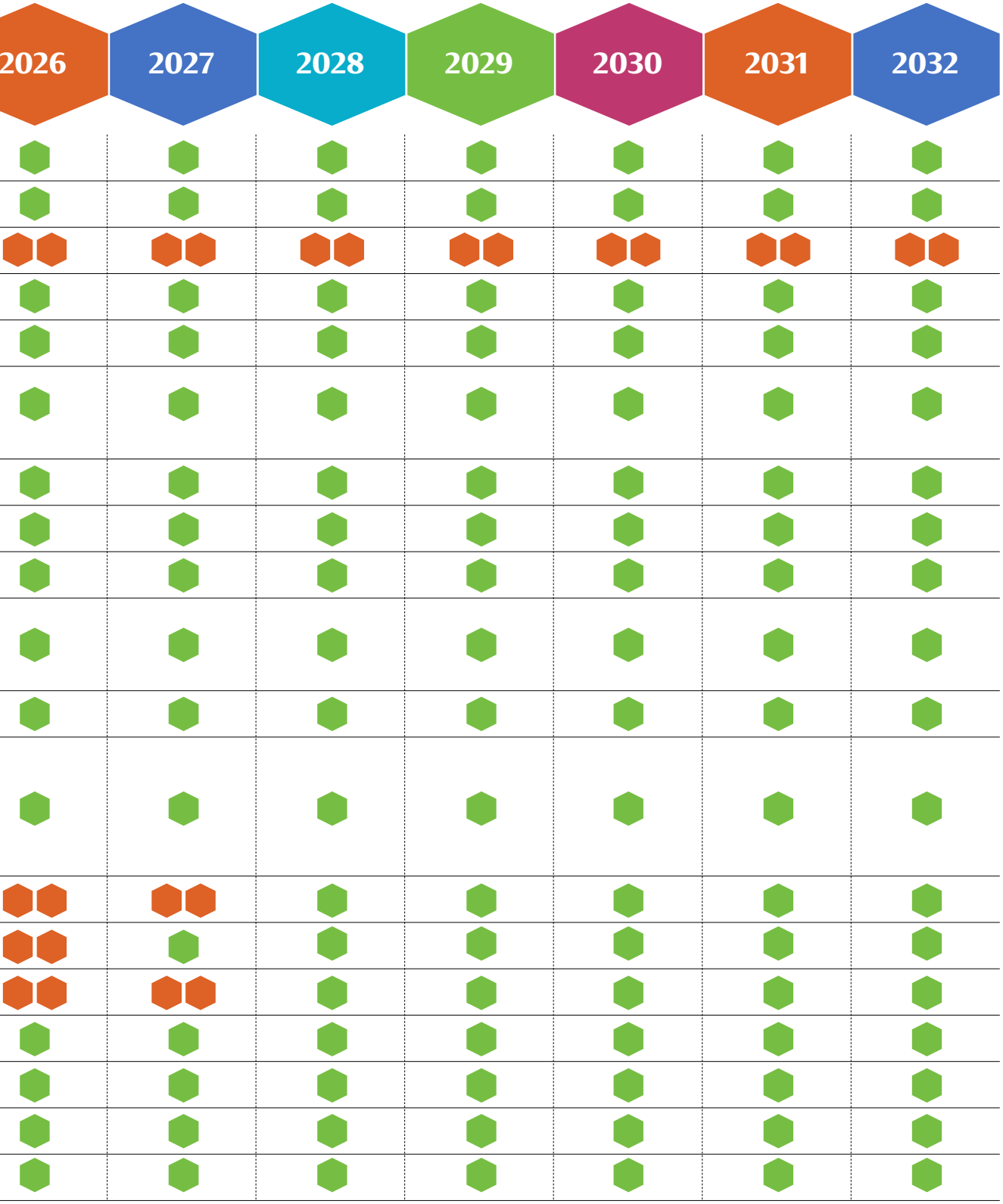
Initiatives resulting in tangible deliverables: digital and ICT assets such as applications, websites, datacentres, systems and networks, and legislation and policies.

# GOALS IMPLEMENTATION TIMELINE

**Workload and focus:**

- High
- Medium
- Low/ ongoing/ maintenance

| | 2023 | 2024 | 2025 |
|---|---|---|---|
| **G1 – Establish the Steering Committee** | Medium | High | Low |
| **G2 – Develop strategies for all Strategic Digital Pillars** | Low | High | Medium |
| **G3 – Develop additional focused strategies** | | Medium | Medium |
| **G4 – Establish the Digital Advisory Team** | High | Medium | Low |
| **G5 – Establish Acceleration Teams** | High | Medium | Low |
| **G6 – Establish a Digital Acceleration and Capabilities Development Centre** | High | Medium | Low |
| **G7 – Fully centralize the government ICT** | Medium | High | Medium |
| **G8 – Centralise ICT spending** | Low | High | Low |
| **G9 – Centralise all government websites and services** | Low | High | High |
| **G10 – Establish a one-stop shop in all sucos: Uma Digital to serve as a government ServiceDesk** | Medium | Medium | Medium |
| **G11 – Enhance national broadband networks** | Low | Low | Low |
| **G12 – Upgrade the government ICT infrastructure, and extend the government network to all government administration, sucos, schools, and health facilities.** | Medium | High | Medium |
| **G13 – Update education curriculums** | | High | Medium |
| **G14 – Fully establish the Unique ID** | Low | High | Medium |
| **G15 – Develop Portal Mekado Bo'ot** | | High | Medium |
| **G16 – Develop a digital and ICT investment strategy** | | High | Medium |
| **G17 – National Digital and ICT Skills development plan** | Medium | High | Low |
| **G18 – Develop enabling legislation and policies** | High | High | Low |
| **G19 – Establish a new Ministry** | Medium | High | Medium |


2026
2027
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2032

# RECOMMENDED NEXT ACTIONS

1. To establish and develop the Steering Committee for Digital and ICT Development, a few actions are recommended to the government:

   I. Country visits to the region, the Pacific and ASEAN, along with CPLP and other nations.

   II. Short video-conferences for members to participate with the objective of hearing from either experts, or other nations, on how particular challenges were tackled, what solutions were and are being implemented for specific sectors and development priorities.

   III. Organise visits to Timor-Leste of experts in varied areas to conduct short workshops with the steering committee members and longer trainings and workshops with the Digital Advisory Team, the Digital Acceleration Teams, the Civil Society and the Private Sector.

2. Develop a sectorial strategy for governance, which includes the following:

   I. Identification of all government services, whether government to government, government to citizens or government to private sector.

   II. Identification of which of the above services are already available online.

   III. A mapping of all IT and information systems in use by the Government of Timor-Leste with the following scope in mind:

      a) Identify use cases of information systems that can benefit from the integration with the Unique Identifier,

      b) Evaluate the feasibility of each system to be integrated with the Unique Identifier, identify the requirements in order to make such systems interoperable with the Unique Identifier.

      c) Assess the quality of the data stored in the identified systems.

      d) Identify new systems that are necessary to develop a digital government.

   IV. Develop an interoperability policy and architecture for the Government of Timor-Leste's systems, as well as the necessary technical documentation, based on the Unique Identifier for interoperability.

   V. A plan for the development and implementation of One-Stop-Shops in each Suco, using digital and information and communication technologies. The one-stop-shop should serve as a ServiceDesk for citizens and link central government, local government, CSOs, Development Partners.

   VI. Plan for the development of a single portal for the Government of Timor-Leste that will serve as an aggregator of all other government portals.

   VII. Plan the development of a common interface for all government external-facing applications.

3. Develop a sectorial strategy to leverage digital and information and communication technologies with the aim of supporting and developing an Inclusive Digital Economy.

4. Develop a sectorial strategy to leverage digital and information and communication technologies with the aim of supporting and developing the Education sector.
5. Develop a sectorial strategy to leverage digital and information and communication technologies with the aim of supporting and developing the Health sector.
6. Develop a sectorial strategy to leverage digital and information and communication technologies with the aim of supporting and developing the Agriculture sector.
7. The Steering Committee should, on an ongoing basis, develop additional sectorial strategies to leverage digital and information and communication technologies with the aim of supporting and developing the additional sectors in alignment with the government priorities.
8. TIC Timor should establish a Digital Advisory Team aimed at supporting the government agencies in the digital transformation journey, particularly to support the digital acceleration teams.
9. Mandate all government agencies to nominate a group of two or three public servants to be responsible for Digital Acceleration in their respective agencies.
10. TIC Timor should work on developing all plans in partnership with government agencies, the digital acceleration teams, the private sector and the civil society, and engage all in the Digital Acceleration and Capabilities Development Centre in training programs, workshops and other initiatives.
11. The Digital Acceleration and Capabilities Development Centre should be a place for designing solutions and implementation plans in partnership with government agencies and the civil society.
12. Establish and develop partnerships for trainings, workshops, experience sharing with other nations, development partners, the private sector and other stakeholders.
13. Develop a plan for the progressive and complete centralization of the government ICT at TIC Timor that fully centralizes all areas of ICT.
14. Develop policies, instruments and mechanisms for the centralised coordination for all government budget, investment and expenditure in digital and information and communication technologies.
15. Initiate the development of a centralised government web portal.
16. Develop a comprehensive National Digital and ICT Skills development plan and Digital and ICT Skills assessment.
17. Develop a plan to enhance national broadband networks and invest in improving the quality and cost of Internet connectivity.
18. Develop a digital, ICT and Telecommunications investment attracting strategy.

# ELEMENTS FOR A SUCCESSFUL IMPLEMENTATION

The success of Timor Digital 2032 will depend on several key factors: maintaining political will, establishing senior coordination, developing a close engagement with civil society and the private sector, being able to adapt and align, taking appropriate coordination mechanisms, having clear implementation plans, and developing skills in digital and ICTs. These factors will ensure a successful implementation and that the strategy remains relevant and useful, answering the needs of government and citizens alike and allowing everyone to add value.

### Political will and coordination

It is crucial to ensure that Timor Digital 2032 is and remains a priority for the entire government and the successive governments. For that, it should be included in the National Strategic Plan, and it is imperative that the Steering Committee for Digital and ICT Development meets regularly, and its attending members remain at the most senior level.

### Civil Society and Public Sector engagement

To ensure that a citizen-centric digital government is developed, the society has the tools and skills to enjoy the benefits of digital and is able to create and add value, and an ecosystem of digital innovation is created, it is important to closely collaborate with the Civil Society and the Private Sector at all levels, from the strategic level down to experimentation and the design of service interfaces.

### Adaptative alignment with government priorities

The Timor Digital 2032 strategic plan establishes the initial Strategic Digital Pillars to leverage the use of digital and ICTs. While this focus is initially in the areas with higher impact, it is critical to progressively extend the strategy to additional pillars and to develop focused strategies for other areas in alignment with the government development priorities. It is also important to revise and publish the Timor Digital 2032 every year.

### Working groups, experimentation, acceleration, and innovation

Foster collaboration by establishing working groups, experimentation, acceleration, and innovation spaces with the participation of the government, civil society organisations, the private sector and other stakeholders to focus on service delivery and back-office improvement, all with the participation of the civil society.

### Skills

Ensure that emphasis is placed on developing the skills of policymakers, technical staff, civil society organisations and citizens so all can drive the digital development of Timor-Leste.

### Implementation plans

Also critical for success is for the GoTL to, after the approval of the strategy to develop detailed implementation plans and roadmaps with tangible deliverables, timelines and the required steps to implement the strategic goals.

Governance
Inclusive Economy
Agriculture
Health
Data Protection and Privacy
Interoperability
Education

Timor Digital 2032

# Timor Digital 2032